AI-Powered Predictive Maintenance for Industrial Equipment

Key Performance Indicators (KPIs)

Metrics for Project Success

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1. Introduction

This document defines the Key Performance Indicators (KPIs) used to measure the success of the AI-Powered Predictive Maintenance project. KPIs span model performance, system reliability, MLOps practices, and project delivery across all three phases.

2. RUL Prediction (LSTM – CMAPSS)

KPI	Target	Measurement Method	Owner
RMSE (Root Mean Square Error)	**< 25 cycles**	MLflow / Test Set Evaluation	Alaa, Rowan
MAE (Mean Absolute Error)	**< 20 cycles**	MLflow / Test Set Evaluation	Alaa, Rowan
R^2 Score	**> 0.80**	MLflow / Test Set Evaluation	Alaa, Rowan
Model Training Time	**< 60 min per run**	MLflow Timing Logs	Alaa
Inference Latency (API)	**< 200 ms per request**	FastAPI Response Logs	Alaa

3. Acoustic Anomaly Detection (Autoencoder – MIMII)

KPI	Target	Measurement Method	Owner
ROC-AUC Score	**> 0.90**	MLflow / Test Set Evaluation	Alaa
F1-Score (Anomaly Class)	**> 0.85**	MLflow / Test Set Evaluation	Alaa
Precision (Anomaly Detection)	**> 0.80**	MLflow / Test Set Evaluation	Alaa
Recall (Anomaly Detection)	**> 0.85**	MLflow / Test Set Evaluation	Alaa
False Positive Rate	**< 10%**	MLflow / Confusion Matrix	Alaa
Reconstruction Error Threshold Accuracy	**> 88%**	MLflow	Alaa

4. MLflow & Experiment Management

KPI	Target	Measurement Method	Owner
Experiment Tracking Coverage	**100% of runs logged**	MLflow UI	Alaa, Nada
Model Registry Versions	**All production models versioned**	MLflow Model Registry	Alaa, Nada
Reproducibility Rate	**100% (same seed = same result)**	Validation Runs	Alaa, Nada

5. Deployment & System

KPI	Target	Measurement Method	Owner
API Uptime	**> 99%**	FastAPI Health Check Endpoint	Alaa
Average Response Time	**< 200 ms**	API Monitoring Logs	Alaa
Model Drift Detection	**Alert within 24 hours of drift**	Monitoring Pipeline	Nada
Retraining Trigger Accuracy	**Pipeline fires correctly on threshold breach**	Automated Pipeline Logs	Nada

6. Project Delivery

KPI	Target	Measurement Method	Owner
Phase 1 Completion (Data)	**By Feb 15, 2025**	Task Tracking	All
Phase 2 Completion (Models)	**By Apr 15, 2025**	Task Tracking	All
Phase 3 Completion (MLOps & Deploy)	**By May 15, 2025**	Task Tracking	All

KPI	Target	Measurement Method	Owner
Documentation Coverage	**All deliverables documented**	Document Repository	All

7. KPI Review Process

KPIs will be reviewed at the end of each project phase and tracked continuously via MLflow dashboards and API monitoring logs:

- Phase 1 Review (Feb 15): Data quality metrics, preprocessing completeness
- Phase 2 Review (Apr 15): All model performance KPIs evaluated against targets
- Phase 3 Review (May 15): Deployment, system uptime, drift detection, and delivery KPIs finalized
- Weekly: Team sync to spot underperforming KPIs early and adjust course